Analysing Smart City Bike Sharing Data using Power BI

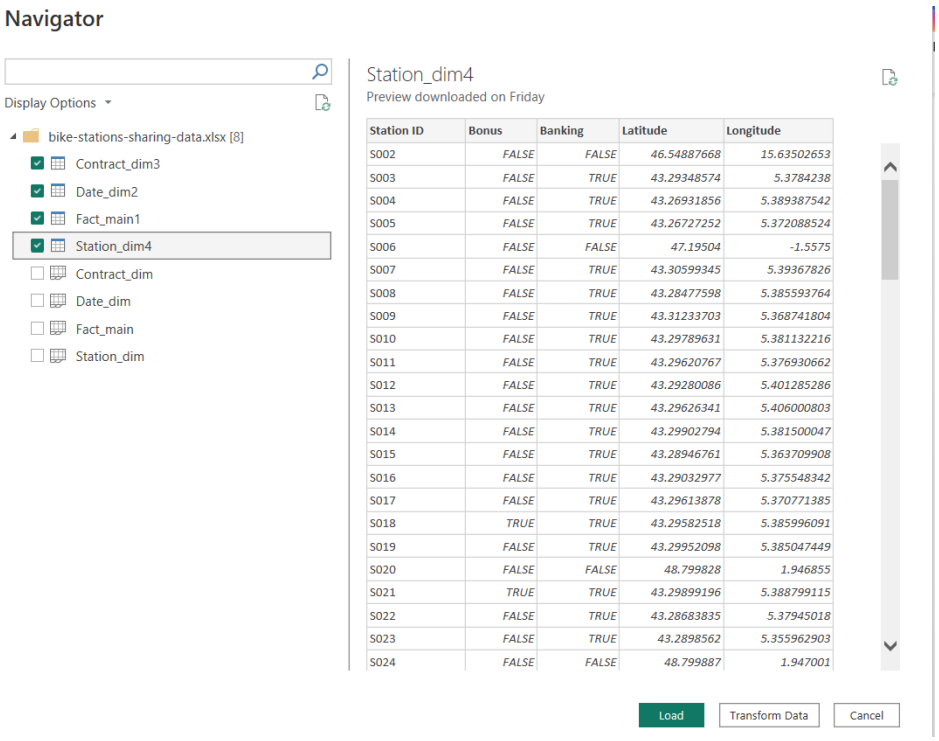
Problem Statement:

Public bike-sharing systems generate continuous data from hundreds of stations across different cities. The task is to analyse this real-time bike station dataset to understand station performance, usage efficiency, and operational patterns across cities.

Applying Power BI skills—from data cleaning and modelling to DAX and dashboarding—to turn raw data into actionable urban mobility insights.

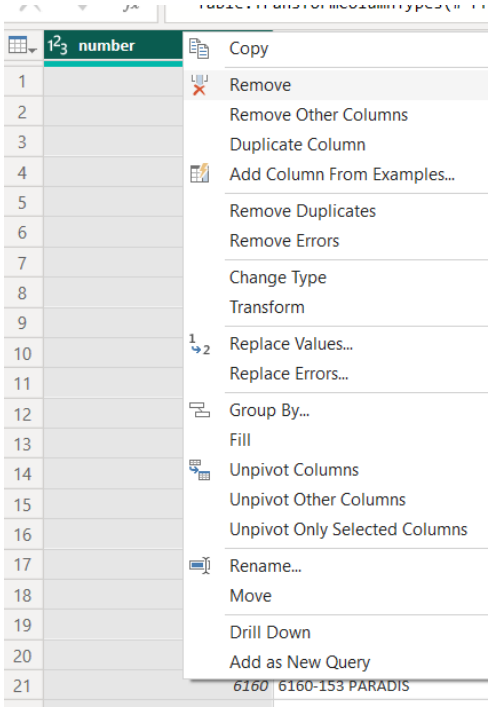
Data Cleaning Process Using Power BI:

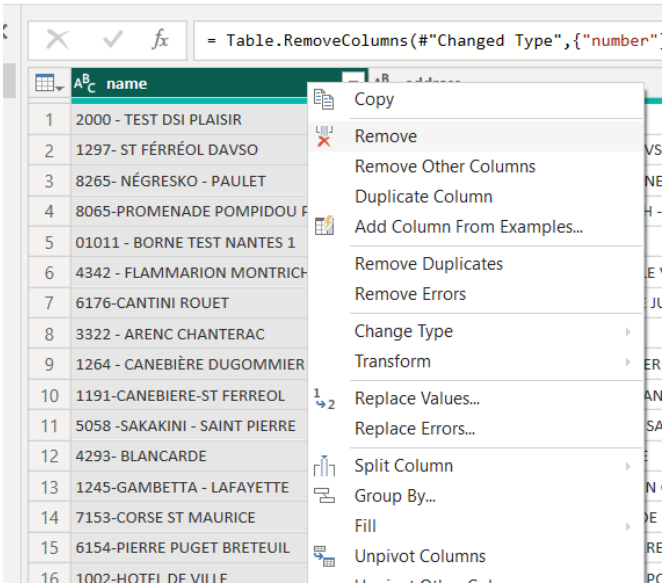
- The Excel file is imported into Power BI

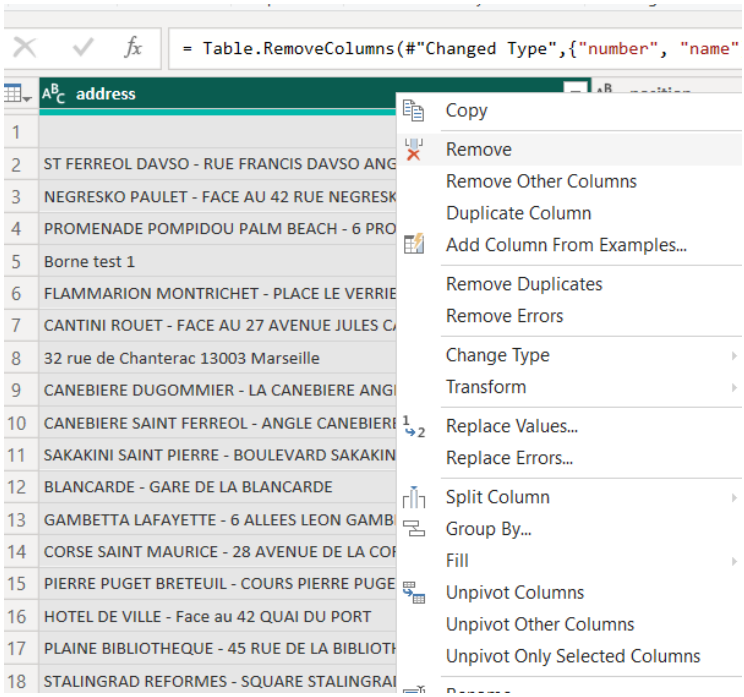


Station ID	Bonus	Banking	Latitude	Longitude
S002	FALSE	FALSE	46.54887668	15.63502653
S003	FALSE	TRUE	43.29348574	5.3784238
S004	FALSE	TRUE	43.26931856	5.389387542
S005	FALSE	TRUE	43.26727252	5.372088524
S006	FALSE	FALSE	47.19504	-1.5575
S007	FALSE	TRUE	43.30599345	5.39367826
S008	FALSE	TRUE	43.28477598	5.385593764
S009	FALSE	TRUE	43.31233703	5.368741804
S010	FALSE	TRUE	43.29789631	5.381132216
S011	FALSE	TRUE	43.29620767	5.376930662
S012	FALSE	TRUE	43.29280086	5.401285286
S013	FALSE	TRUE	43.29626341	5.406000803
S014	FALSE	TRUE	43.29902794	5.381500047
S015	FALSE	TRUE	43.28946761	5.363709908
S016	FALSE	TRUE	43.29032977	5.375548342
S017	FALSE	TRUE	43.29613878	5.370771385
S018	TRUE	TRUE	43.29582518	5.385996091
S019	FALSE	TRUE	43.29952098	5.385047449
S020	FALSE	FALSE	48.799828	1.946855
S021	TRUE	TRUE	43.29899196	5.388799115
S022	FALSE	TRUE	43.28683835	5.37945018
S023	FALSE	TRUE	43.2898562	5.355962903
S024	FALSE	FALSE	48.799887	1.947001

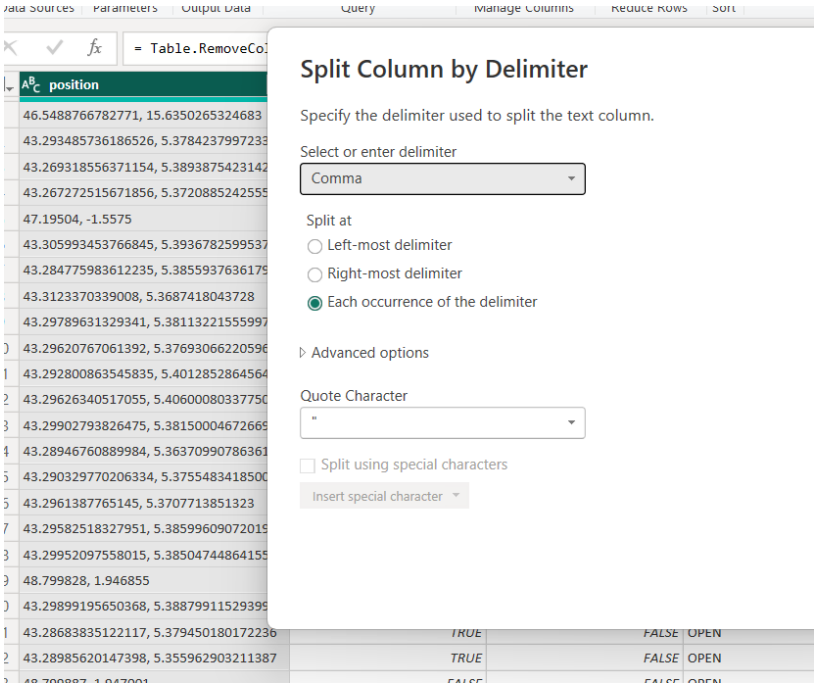
- Unnecessary columns were removed to enhance the quality and efficiency of the analysis.



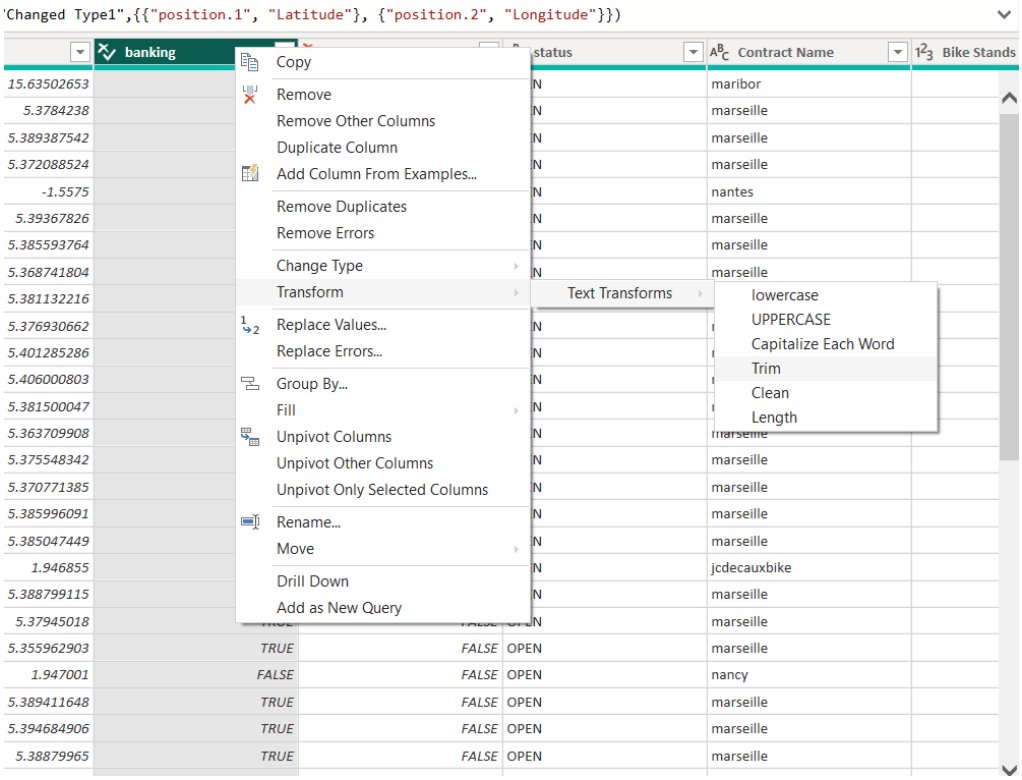




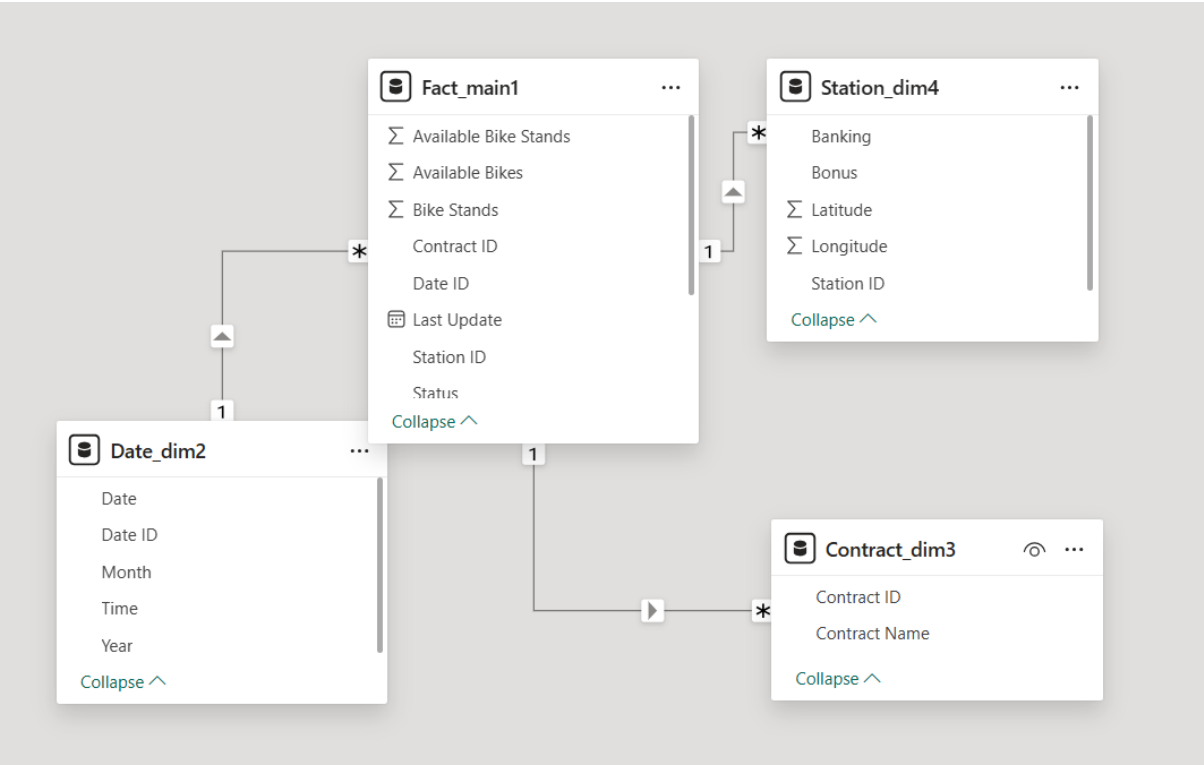
- The position column was split into latitude and longitude to improve data clarity and enable location-based analysis.



- Data was standardized using functions such as TRIM, CLEAN, and PROPER to ensure consistency and accuracy.



Data Modelling:



Overall Model Structure

- It contains 1 Fact Table → Fact_main1
- And 3 Dimension Tables:
  - Date_dim2
  - Station_dim4
  - Contract_dim3

This is a Star Schema.

Fact Table:

Fact_main1

- Contains:
- Available Bikes
- Available Bike Stands
- Bike Stands
- Contract ID

- Date ID
- Station ID
- Last Update
- Status

Station ID	Date ID	Status	Contract ID	Bike Stands	Available Bike Stands	Available Bikes	Last Update
S009	D008	Open	C009	20	13	26	14-09-2022 13:27:09
S011	D010	Open	C011	20	2	4	03-10-2022 11:34:01
S036	D035	Open	C036	20	20	43	21-03-2023 20:14:27
S039	D038	Open	C039	20	20	13	21-11-2022 15:22:40
S044	D043	Open	C044	20	20	38	14-11-2023 13:21:56
S057	D056	Open	C057	20	8	46	20-05-2024 11:12:54
S079	D078	Open	C079	20	220	23	18-12-2023 17:04:43
S086	D085	Open	C086	20	20	43	03-09-2024 11:14:57
S088	D087	Open	C088	20	16	31	09-06-2025 13:11:37
S094	D093	Open	C094	20	120	18	11-06-2025 12:34:42
S096	D095	Open	C096	20	18	25	28-01-2025 11:32:28
S102	D101	Open	C102	20	20	49	25-03-2025 13:55:04
S107	D106	Open	C107	20	20	54	22-04-2025 11:38:11
S111	D110	Open	C111	20	20	29	28-05-2025 11:11:35
S113	D112	Open	C113	20	220	43	03-06-2025 11:22:35
S115	D114	Open	C115	20	5	7	05-06-2025 11:43:24
S116	D115	Open	C116	20	20	41	06-06-2025 13:36:37
S119	D118	Open	C119	20	19	14	27-06-2025 14:07:46
S131	D130	Open	C131	20	20	51	07-10-2025 10:53:43
S134	D133	Open	C134	20	220	5	23-10-2025 10:04:38
S138	D137	Open	C138	20	120	44	23-06-2025 11:29:36
S148	D147	Open	C148	20	18	47	08-07-2022 12:29:34
S157	D156	Open	C157	20	20	35	28-09-2022 16:52:17
S164	D163	Open	C164	20	16	43	10-10-2022 11:17:35
S203	D202	Open	C203	20	17	8	20-05-2024 11:15:21
S207	D206	Open	C207	20	20	51	28-02-2024 14:31:53
S218	D217	Open	C218	20	20	15	03-09-2024 11:13:36
S221	D220	Open	C221	20	4	36	18-11-2024 11:16:15

Dimension Tables

Date_dim2

Contains:

- Date ID
- Date
- Month
- Year
- Time

Date ID	Date	Month	Year	Time
D418	12	11	2025	12:28:30
D419	12	11	2025	14:24:16
D420	12	11	2025	13:25:58
D421	12	11	2025	14:12:58
D422	12	11	2025	14:47:35
D423	12	11	2025	14:45:10
D424	12	11	2025	14:47:57
D425	12	11	2025	14:48:49
D426	12	11	2025	14:48:34
D427	12	11	2025	14:48:01
D428	12	11	2025	14:47:54
D429	12	11	2025	14:48:01
D430	12	11	2025	14:47:50
D431	12	11	2025	14:49:10
D432	12	11	2025	14:47:57
D433	12	11	2025	14:49:32
D434	12	11	2025	14:48:17
D435	12	11	2025	14:47:52
D436	12	11	2025	14:49:16
D437	12	11	2025	14:48:00
D438	12	11	2025	14:48:06
D439	12	11	2025	14:48:42
D440	12	11	2025	14:48:58
D441	12	11	2025	14:48:06
D442	12	11	2025	14:49:00
D443	12	11	2025	14:47:58
D444	12	11	2025	14:48:46

Station_dim4

Contains:

- Station ID
- Latitude
- Longitude
- Banking
- Bonus

Station ID	Bonus	Banking	Latitude	Longitude
S002	False	False	46.5488766782771	15.6350265324683
S006	False	False	47.19504	-1.5575
S020	False	False	48.799828	1.946855
S024	False	False	48.799887	1.947001
S044	False	False	45.748286	4.896975
S045	False	False	43.240392	5.362533
S060	False	False	37.3999401269442	-5.92516396433934
S062	False	False	48.7998093181557	1.94670385760583
S063	False	False	43.6092342351654	1.42652571179497
S065	False	False	47.736212	7.31875
S067	False	False	50.803358	4.373483
S073	False	False	47.751456	7.33392
S076	False	False	47.743132	7.352147
S080	False	False	47.750923	7.329854
S082	False	False	47.7526919378613	7.34645885817564
S083	False	False	45.755068	4.871457
S084	False	False	37.39074021	-5.959164841
S085	False	False	48.799648	1.945439
S086	False	False	37.4178556017479	-5.96929614545535
S087	False	False	50.859212	4.34551
S088	False	False	49.901646	2.303244
S089	False	False	45.752923	4.898891
S090	False	False	45.780403	4.904954
S091	False	False	49.8922382866591	2.30769408025555
S092	False	False	47.7500690513047	7.33871413913301
S093	False	False	50.867643	4.385817
S094	False	False	49.890015514268	2.29122726421607

Contract_dim3

Contains:

- Contract ID
- Contract Name

Contract ID	Contract Name
C002	Maribor
C003	Marseille
C004	Nantes
C005	Jcdecauxbike
C006	Nancy
C007	Rouen
C008	Lyon
C009	Santander
C010	Creteil
C011	Seville
C012	Besancon
C013	Namur
C014	Mulhouse
C015	Bruxelles-Capitale
C016	Amiens
C017	Cergy-Pontoise
C018	Toulouse
C019	Luxembourg
C020	Lillestrom
C021	Valence
C022	Dublin
C023	Vilnius
C024	Toyama
C025	Lund
C026	Ljubljana

Relationships:

- Date_dim2 (1) → Fact_main1 (*)
- Station_dim4 (1) → Fact_main1 (*)
- Contract_dim3 (1) → Fact_main1 (*)
- This is Many-to-One (*:1)

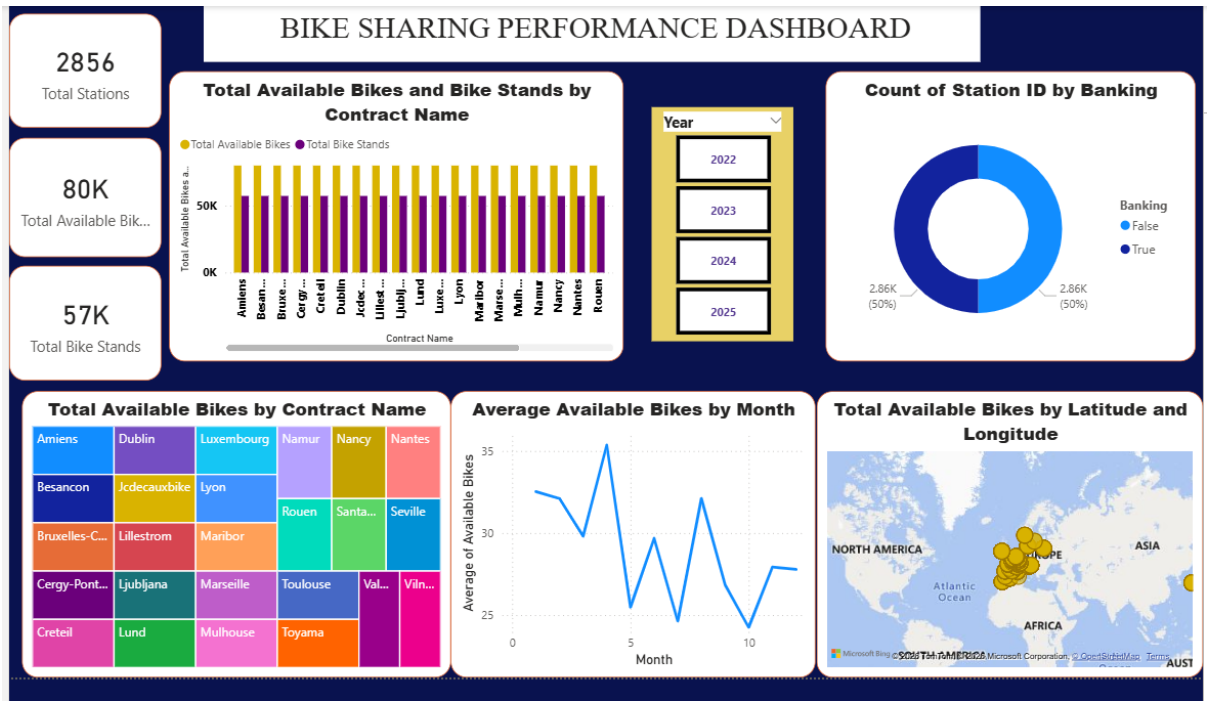
DAX Measures:

Total Stations = DISTINCTCOUNT(Fact_main1[Station ID])

Total Available Bikes = SUM(Fact_main1[Available Bikes])

Total Bike Stands = SUM(Fact_main1[Bike Stands])

Final Dashboard:



1. Contract-wise Performance Analysis:

- The comparison between Total Available Bikes and Total Bike Stands across contracts shows that most contracts have high capacity but relatively similar availability levels.
- This indicates that capacity is not the limiting factor, but rather usage and distribution efficiency.
- Some contracts slightly outperform others, but there is no extreme dominance, suggesting a well-distributed system overall.
- However, the gap between bike stands and available bikes highlights that a significant portion of infrastructure is not fully utilized.

Business Insight:

Focus should shift from adding capacity to optimizing bike allocation and redistribution.

2. Time-based Trend Analysis:

- The average available bikes fluctuates across months, with noticeable peaks and drops.
- Certain months show higher availability, which implies lower demand or excess supply.
- Other months show sharp declines, indicating high usage periods (peak demand).
- The pattern is not stable, meaning demand is seasonal and inconsistent.

Business Insight:

- Introduce dynamic rebalancing strategies based on seasonal demand.
- High-demand months require more bike allocation, while low-demand periods need capacity optimization.

3. Feature Impact Analysis:

- The banking vs non-banking distribution is almost equal (50%-50%), indicating no heavy skew toward either category.
- This suggests that the system is evenly designed in terms of infrastructure features.
- However, since availability is not drastically different, banking may not be a strong driver of usage.

Business Insight:

- Investment in banking features should be evaluated carefully, as it may not significantly impact performance.
- Further analysis can be done to see if banking affects specific locations or peak usage scenarios.

4. Geographic & Distribution Analysis:

- The map shows a strong concentration of bike stations in specific regions (primarily Europe).
- This indicates that the system is regionally concentrated rather than globally distributed.
- The tree map shows that bike availability is spread across many contracts, with no single contract dominating excessively.
- However, some regions may have higher density but not necessarily higher utilization, suggesting imbalance in distribution.

Business Insight:

- There is an opportunity to expand into underrepresented regions.
- Existing regions may require better balancing of bikes across stations rather than increasing total numbers.

Conclusion:

Overall, the analysis reveals that although the system has adequate infrastructure, inconsistent demand patterns and underutilized capacity indicate the need for improved bike allocation, dynamic rebalancing, and data-driven operational strategies.